$*$ - asterisk

UNIT-5

Binary Operations :

Let A be any non-empty set. Then a symbol $*$ is called a binary operation on A, if A is closed under $*$ i.e $a, b \in A$

$\Rightarrow a * b \in A, \forall a, b \in A$

For Ex: Consider the set of natural numbers

$\mathbb{N} = \{1, 2, 3, \ldots\}$

Then usual addition and multiplication are binary operations on $\mathbb{N}$ because $a, b \in \mathbb{N} \Rightarrow a+b \in \mathbb{N}$ and $a, b \in \mathbb{N} \Rightarrow a \times b \in \mathbb{N}$,

But substraction and division are not binary operations on $\mathbb{N}$ because $3, 4 \in \mathbb{N}$, but $3-4 \notin \mathbb{N}$ and $2, 3 \in \mathbb{N}$ but $\frac{2}{3} \notin \mathbb{N}$

$*$ Binary operations are generally denoted by $*, +, \circ \ldots$ etc

Algebraic Structure :

A non-empty set A together with a binary operation $*$ is called an algebraic structure and it is written as $(A, *)$

Groupoid :

Let G be a non-empty set and $*$ be a binary operation on G. Then the algebraic structure $(G, *)$ is called a groupoid if the operation $*$ satisfies closure property.

G_1 (closure) : $a, b \in G$

$\Rightarrow a * b \in G$

Semi-Group :

Let G be a non-empty set and $*$ be a binary operation on G. Then algebraic structure $(G, *)$ is called semi-group if the operation $*$ satisfies closure and associative property.

(i) G_1 (closure) : $a, b \in G$

$\Rightarrow a * b \in G$

(ii) G_2 (Associative) : $a * (b * c) = (a * b) * c \quad \forall a, b, c \in G$

Monoid :

Let G be a non-empty set and $*$ be a binary operation on G. Then the algebraic structure $(G, *)$ is called a monoid, if the following properties are satisfied.

(i) Closure (G_1) : $\forall a, b \in G \Rightarrow a * b \in G$

(ii) G_2 - Associativity : $a * (b * c) = (a * b) * c$

$\forall a, b, c \in G$

(iii) G_3 (Identity) : There exists and element $e \in G$ such that $a * e = e * a = a \; \forall a \in G$.

Group :

Let G be a non-empty set on which a binary operation $*$ is defined. Then G is called a group if the following four properties are satisfied.

a) Closure properties : G_1

$a, b \in G \Rightarrow a * b \in G$

b) Associative properties (G_2)

$a, b, c \in G \Rightarrow a * (b * c) = (a * b) * c$

c) Existence of Identify (G3):
There exists an elements $e \in G$ such that $a * e = a = e * a \quad \forall a \in G$, e is called identity element for $*$ in G.

d) Existence of Inverse (G4):
For every $a * G$, there exists an element $b \in G$ such that $a * b = e = b * a$
Here b is called the inverse of a and is written as $b = a^{-1}$

Note: (1) If $*$ is usual addition, then $e = 0$ and $b = -a$
(2) If $*$ is usual multication, then $e = 1$, $b = \frac{1}{a}$

Abelian / Commutative Group:
A group G is said to be abelian if $a, b \in G \Rightarrow a * b = b * a$

Finite Group: A group G is said to be finite if it contains finite number of elements, otherwise it is called infinite group.

Order of a group: The number of elements in finite group G is called order of the group & is denoted by $O(G)$.

Examples :

(1) $\mathbb{Z} = \{0, \pm 1, \pm 2, \cdots\}$ – set of integers

– $\mathbb{Z}$ is an abelian group under usual addition because ~~of~~ all the properties G_1, G_2, G_3, G_4 hold. also $a+b = b+a \quad \forall a, b \in \mathbb{Z}$

Also, $\mathbb{Z}$ is not a group under usual multiplication because G_4 does not hold. as $3 \in \mathbb{Z}$ but $\frac{1}{3} \notin \mathbb{Z}$.

(2) $\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$ (Rational Numbers)

(3) $\mathbb{R} = \{0, 1, 2, 3, \cdots\}$ (Real Number)

(4) $\mathbb{C} = \{a + ib, a, b \in \mathbb{R}, i = \sqrt{-1}\}$ (complex numbers)

Here $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are abelian groups under usual addition but they are not groups under usual multiplication.

* If we consider the set of non-zero $\mathbb{Q}, \mathbb{Z}, \mathbb{R}, \mathbb{C}$. then they are all abelian groups under usual multiplication., but they are not group under usual ~~addition~~ multiplication

(5) $G = \{0, 1, -1\}$ $\rightarrow$ Not a group under usual addition as well as usual multiplication

(6) $G = \{e\}$ $\rightarrow$ group of order 1 under multiplication

(7) $G = \{1, -1\}$ – finite group of order 2 under multiplication

Q. prove that the set $\{1, 2, 3, 4, 5, 6\}$ of order 6 is finite abelian group under multiplication module 7 as composition.

$\Rightarrow$ $A = \{1, 2, 3, 4, 5, 6\}$

To prove: $(A, \times_7)$ is abelian group

we verify G_1, G_2, G_3, G_4, G_5

$\times_7$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

identity element

(Inverse elements are present)

G_1: From the above table, we observe that, $a \cdot b \in A$ $\forall a, b \in A$

$\therefore$ G_1 is verified.

G_2: We observe that,

$1 \cdot (2 \cdot 3) = (1 \cdot 2) \cdot 3$

$\Rightarrow$ $1 \cdot 6 = 2 \cdot 3$

$6 = 6$

$\therefore$ $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ $\forall a, b \in A$

G_3: Here $1 \in A$ and $a \cdot 1 = 1 \cdot a = a$

$\therefore$ 1 is the identity element

$\therefore$ G_3 is verified.

G_4 : From the table, we observe that

$2 \cdot 4 = 1 = 4 \cdot 2$

$3 \cdot 5 = 1 = 5 \cdot 3$ Where 1 is the identity element.

$\therefore a \cdot b = 1 = b \cdot a \quad \forall a, b \in A$, where b is the multiplicative inverse of a.

$\therefore$ Multiplicative inverse of each element of A exists

$\therefore G_4$ is verified.

G_5 : From the table we observe that.

$a \cdot b = b \cdot a \quad \forall a, b \in A$

$\therefore G_5$ is verified.

$(A, \times_5)$ is verified abelian group.

(ii) To prove: Every $a \in G$ has a unique inverse in G

Let G be a group with identity e

and let $a \in G$.

$\therefore$ Inverse of a exists in G (by G_4)

Suppose, x and y are the inverses of a in G.

$$\left.\begin{aligned} a \cdot x &= e = x \cdot a \\ a \cdot y &= e = y \cdot a \end{aligned}\right\} \longrightarrow (1)$$

Since $a, x, y \in G$ and G is a group

$$\therefore \quad x(a \cdot y) = (x \cdot a) \cdot y \qquad \longrightarrow \text{by } G_2$$

$$x \cdot e = e \cdot y$$

$$x = y$$

$\therefore$ every $a \in G$ has a unique inverse in G.

(iii) To prove: For every $a \in G$, $(a^{-1})^{-1} = a$

Let G be a group with identity e

and let $a \in G$

$\Rightarrow \quad a^{-1} \in G$ s.t $a \cdot a^{-1} = e = a^{-1} \cdot a \longrightarrow (2)$

consider,

$$a \cdot a^{-1} = e$$

Multiply both sides by $(a^{-1})^{-1}$

$$a \cdot a^{-1} (a^{-1})^{-1} = e \cdot (a^{-1})^{-1}$$

$$a [a^{-1} (a^{-1})^{-1}] = (a^{-1})^{-1}$$

$$a \cdot e = (a^{-1})^{-1}$$

$$a = (a^{-1})^{-1} \qquad \forall a \in G.$$

(iv) For all $a, b \in G$, $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$

Let G be a group with identity e.

& let $a \cdot b \in G$

$$a^{-1} b^{-1} \in G \qquad \longrightarrow \text{by } G_4$$

$$b^{-1} a^{-1} \in G \qquad \longrightarrow \text{by } G_1$$

Cancellation Laws:

Let a, b, c be the elements of a group G then

(i) $ab = ac \Rightarrow b = c$ (left cancellation law)

(ii) $b \cdot a = c \cdot a \Rightarrow b = c$ (Right cancellation law)

Note: If no positive power of g is equals to e, then g is said to be infinite (or zero) order

Order of an element:

Let $(G, *)$ be a group and $a \in G$, then the least/smallest positive integer n (if it exist) such that $a^n = e$, then is called the order of a, $a \in G$. Here e is the identity element of G

The order of an element $a \in G$ is denoted by $o(a)$

$\boxed{\text{Note:}}$ If $o(a) = n$ then $a^n = e$

Ex: Let $G = \{1, -1, i, -i\}$ be the group w.r.t multiplication. Here 1 is the identity element. then the order of each element

Element	order	Reason
1	$o(1) = 1$	$1^1 = 1$
-1	$o(-1) = 2$	$(-1)^2 = 1$
i	$o(i) = 4$	$i^4 = 1$
$-i$	$o(-i) = 4$	$(-i)^4 = 1$

Ex: Find the order of each element of the given $G = \{0, 1, 2, 3\}$ with addition modulo 4.

$\Rightarrow$ $G = \{0, 1, 2, 3\}$

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Here 0 is the identity element

element	order	Reason
0	$o(0) = 1$	$0^1 = 0$
1	$o(1) = 4$	$1^4 = 0 \mid = 0$
2	$o(2) = 2$	$2^2 = 4 = 0$
3	$o(3) = 4$	$3^4 = 12 = 0$

Cosets :

Let H be a subgroup of a group G and $a \in G$, then

(i) The set $H * a = \{h * a \mid h \in H\}$ is called the right coset of H in G determined by a

(ii) The set $a * H = \{a * h \mid h \in H\}$ is called the left coset of H in G determined by a

Note :

1. In addition notation, the above notation definition takes the form $H + a = \{h + a \mid h \in H\}$ and $a + H = \{a + h \mid h \in H\}$.

2. If e is the identity element of G, then we have $e * H = H = H * e$
Hence H is itself is a right as well as left coset determined by e.

3. In general $a * H \neq H * a$
But if G is an abelian group, then $a * H = H * a$

Ex: Let $G = \{0, \pm 1, \pm 2, \pm 3, \cdots\}$ and $H = \{\cdots -9, -6, -3, 0, 3, 6, 9, \cdots\}$

Find left and right cosets.

Soln: Then $H + 0 = H$

$H + 1 = \{\cdots -8, -5, -2, 1, 4, 7, 10, \cdots\}$

$H + 2 = \{\cdots -7, -4, -1, 2, 5, 8, 11, \cdots\}$

$H + 3 = \{\cdots -6, -3, 0, 3, 6, 9 \cdots\} = H$

$H + 4 = \{\cdots -5, -2, 1, 4, 7, 10, 13, \cdots\} = H + 1$

Similarly, $H + 5 = H + 2$.

$H + 6 = H + 3$ and so on

Hence the right as well as left cosets are $H, H+1, H+2$.

Results:

✓ 1. If $(H, *)$ is a subgroup of $(G, *)$ then $a*H = H$ if and only if $a \in H$.

✓ 2. If $(H, *)$ is a subgroup of $(G, *)$ then $a*H = b*H$ if and only if $a^{-1} * b \in H$

✓ 3. If $(H, *)$ is a subgroup of $(G, *)$ then any two left cosets of H are either identical or disjoint

4. There exist a one-one correspondence between the elements of subgroups H and those of any cosets of H is G.

5. If $a*H$ and $b*H$ are two distinct left cosets of H in G, then there exist one-one correspondence between $a*H$ and $b*H$

Lagrange's Theorem:

Statement: The order of any subgroup of a finite group divides the order of the group

⇒ Let $(G, *)$ be a finite group of order n.
& $(H, *)$ be a sub-group of G of order m.
We decompose the set G into a union of a finite number of distinct left cosets of H (say k)

Let,
$a_1H, a_2H, \dots a_kH$ denote the k distinct left cosets of H in G such that

$$G = (a_1 * H) \cup (a_2 * H) \cup \dots \cup (a_k * H)$$

Where, all the k left cosets appearing on the

right are disjoint

$$o(G) = o(a_1 * H) + o(a_2 * H) + \cdots + o(a_k * H)$$

$$n = \underbrace{m + m + \cdots + m}_{(k\text{-times})}$$

$$n = km$$

$$m \mid n$$

$$o(H) \mid o(G)$$

This proves the theorem.

Normal Subgroup:

A subgroup N of a group G is said to be a normal subgroup iff, $gng^{-1} \in N \quad \forall n \in N, \forall g \in G$

Equivalently,

$$gNg^{-1} \subseteq N \quad \forall g \in G$$

Where, $gNg^{-1} = \{gng^{-1} \mid n \in N\}$

i.e. $x \in gNg^{-1} \iff x = gng^{-1}$

Quotient Group / Factor Group

Let N be a normal subgroup of a group G. Then G/N denotes the set of all right cosets or left cosets of N in G.

i.e $G/N = \{Na \mid a \in G\}$ or $\{aN \mid a \in G\}$

i.e $x \in G/N \iff x = Na$ or $x = aN$, for some $a \in G$

Here G/N is called quotient group of factor group.

Homomorphism of Groups:

Let G and G' be two groups. A function $f: G \longrightarrow G'$ is said to be group homomorphism if $f(ab) = f(a) \cdot f(b) \quad \forall a, b \in G$.

$f(a * b) = f(a) \circ f(b)$.

Isomorphism of Groups:

A homomorphism $f: G \longrightarrow G'$ is said to be an isomorphism if f is one-one i.e $f(a) = f(b) \Rightarrow a = b$

Isomorphic Groups:

Two groups G and G' are said to be isomorphic if there exist a homomorphism $f: G \longrightarrow G'$ which is one-one and onto is written as $G \approx G'$

Onto: $f: G \longrightarrow G'$ is onto, then for every $y \in G'$ there exists some $x \in G$ such that $f(x) = y$

Subgroup:

Let $(G, *)$ be a group and H be a non-empty subset of G, then H is called a subgroup of G if H is a group under the operation $*$.

For example:

(1) Consider, $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$, $H = \{0, \pm 2, \pm 4, \dots\}$

Here $H \subset \mathbb{Z}$

Also both H and $\mathbb{Z}$ are groups under the operation addition.

$\therefore$ H is a subgroup of $\mathbb{Z}$.

(2) $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$, $H = \{0, \pm 1\}$

Here $H \subset \mathbb{Z}$

Also $\mathbb{Z}$ is a group under addition but H is not a group under addition as G_1 does not hold in H because $1, 1 \in H$ but $1 + 1 = 2 \notin H$

$\therefore$ H is not a subgroup of G.

Notes:

1. If e is the identity element of a group G then both the set $\{e\}$ and G are subgroups of G. These subgroups $\{e\}$ and G are called trivial subgroups and the other subgroups are called non-trivial subgro

2. If $H \subseteq G$ then associativity property in H obviously holds because $a, b, c \in H$.

$\Rightarrow a, b, c$

$\Rightarrow G$

But G is group

$\therefore$ $a * (b * c) = (a * b) * c$

3. If H is a subgroup of G, then
(a) Identity of H = Identity of G.
(2) Inverse of $a \in H$ = Inverse of $a \in G$.

Ex: Let $G = \{\ldots 3^{-2}, 3^{-1}, 1, 3, 3^2, \ldots\}$ be the multiplicative group consisting of all p. integral power of 3. Let $H = \{1, 3, 3^2, \ldots\}$. Then $H \subset G$ and H is closed with respect to multiplication, But H is not a subgroup of G since the inverse of 3 i.e 3^{-1} or $\frac{1}{3}$ does not belongs to H

Ex. Let G be the additive group of all integers & H be the subset of G consisting of all positive integers Then H is closed with respect to addition. But H is not a subgroup of G since the identity of $0 \notin H$